

Katherina Dayeon

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EDUCATION

University of Central Florida, Burnett Honors College

Bachelor of Science in Computer Science, GPA: 3.8

Orlando, FL

Expected May 2028

EXPERIENCE

Software Engineer Intern | *React, Three.js, Blender, JavaScript, HTML, CSS*

December 2025

Institute for Simulation and Training - UCF

Orlando, FL

- Contributed to a **U.S. Department of Energy** & ORTECC research initiative, creating a solution for smart factories.
- Built an interactive 3D environment using **Three.js** and simulated data to visualize machines and factory conditions.
- Integrated **OpenAI**-powered assistance and machine data to help users interpret system behavior and operational context.
- Improved response time by **30%** by optimizing data flow between the frontend and 3D environment, enhancing usability.

Website & App Developer | *HTML, CSS, Wordpress, Athena, Microsoft Power Apps* August – December 2025

UCF Arboretum

Orlando, FL

- Collaborated within a team of **5** to plan, develop, and implement internal applications to streamline operations.
- Increased website traffic by **20%** by redesigning key pages and resolving critical usability issues.
- Developed user-focused interface designs using **Figma** and **HTML**, improving layout structure and visual appeal.
- Integrated **accessibility** best practices into web design, improving user experience for all visitors.

Design Director | *Figma, Notion, Canva, Adobe Illustrator, Photoshop, Procreate*

Jan – December 2025

Knight Hacks

Orlando, FL

- Managed a team of **20+** people using **Notion** and delegated various project tasks involving hackathon needs.
- Oversaw and refined **10–15** design assets per month leveraging **Adobe Illustrator**, Photoshop, Procreate, and Canva.
- Revamped website and social media, reaching a following of **2000+** on Instagram and increasing follower count by **45%**.
- Rebranded merch and club websites to be used by **6000+** members, increasing member outreach by **50%**.

Software Engineer Intern | *Unity, C#, Blender*

Jan – April 2025

ZuLeris Interactive

Orlando, FL

- Developed an electromagnetic warfare training simulation for the **U.S. Department of Defense** using **Unity**.
- Optimized UI/UX by designing intuitive layouts and improving user flows, increasing task completion efficiency by **15%**.
- Engineered realistic and interactive communication scenarios to improve **10+** users' proficiency and engagement.
- Collaborated with a team to increase soldiers' adaptability in complex electromagnetic environments.

PROJECTS

SMARTLook | *OpenAI, Three.js, REST APIs, React, JavaScript*

December 2025

- Produced a web-based maintenance tool using **React** and **Three.js**, visualizing real-time machine data in a 3D space.
- Integrated **OpenAI** via REST APIs to provide AI assistance, system explanations, and user guidance within the platform.
- Accelerated 3D rendering performance from **10s** to under **4s** by applying DRACO compression and reducing data transfer.

Potara | *Gemini, REST APIs, Express.js, MinIO, PostgreSQL, Prisma, Docker, React, JavaScript*

September 2025

- Developed an art tool powered by **Google Gemini**, providing live AI feedback and ability to draw on a web-based canvas.
- Designed and developed the frontend of the website, utilizing tools like Figma, **TailwindCSS**, and React.
- Reduced live feedback latency from **15s** to under **5s** by optimizing data transfer, using base64-encoded data.

ShapeShift | *React, JavaScript, TypeScript, Figma*

April 2025

- Participated in **Bitcamp 2025**, creating a full stack web app that converts 2D assets into production-ready 3D models.
- Designed and implemented the **React** based frontend of a dev tool web app that converts SVGs into 3D models.
- Developed a responsive UI/UX using React and **Figma**, increasing user engagement by **40%**.

AiSight | *Python, TypeScript, React Native, Figma*

January 2025 – April 2025

- Placed **4th** out of 15 in Knight Hacks Project Launch with accessibility glasses and app for low-visibility users.
- Engineered a **React Native** app to assist the visually impaired, enhancing accessibility through various commands.
- Collaborated with a team of **6**, coordinating frontend and backend development for efficient task distribution.

TECHNICAL SKILLS

Languages: HTML, CSS, Python, C, JavaScript, Java, TypeScript, C#

Frameworks: React, React Native, Unity, Tailwind, Flutter, TKinter, Three.js

Developer Tools: Git, GitHub, Visual Studio Code, Visual Studio, Figma, Blender, Canva, Adobe Illustrator